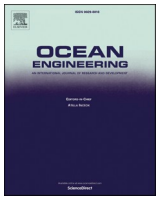
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Timing of stand-on ship actions for cooperative collision avoidance under calm water condition

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ABSTRACT

Rule 17 of the COLREGS states that a stand-on ship should take cooperative action to avoid a collision when the action of the give-way ship will not be sufficient to avoid a collision. However, most studies on collision avoidance have focused on the action of the give-way ship and not on what the stand-on ship should do. In this study, a novel criterion was developed for assessing when a stand-on ship should take action for collision avoidance that is called the computed distance at collision (CDC). Simulations were performed of the maneuvering motions of two ships in various situations to assess whether or not a collision would occur. The results of the collision assessment were assessed by the CDC, which is calculated from the ship's parameters such as the ship's heading, length, and collision angle between the two ships. Meanwhile, the actual distance (AD) between the two ships can be calculated according to their coordinates. When the CDC is greater than the AD, this means that a collision is inevitable. In other words, the stand-on ship should take action to avoid a collision while the CDC is still less than the AD. Of course, this study may have the following limitations: The study was conducted on the same model ship under calm water conditions. The interactions of hull-propeller-rudder-engine of ship and interaction of hull-hull between the two ships was not considered, and uncertainty of MMG (Maneuvering Modeling Group) model may exist. However, this study should help determine the timing of actions by the stand-on ship for collision avoidance, whether manned or unmanned.

1. Introduction

Ships do not have brakes like cars, so the speed and maneuvering are difficult to control because of their operation and size. Thus, two ships can face a situation where they cannot avoid a collision. Ship collisions cause great economic losses (Heij et al., 2011). To ensure navigational safety, the International Maritime Organization (IMO) enacted the Convention on the International Regulations for Preventing Collisions at Sea (COLREGS) in 1972, which applies to all ships in all waters. However, ship collisions still occur despite the introduction and subsequent amendments of COLREGS because the complexity of the marine environment makes it difficult to apply these regulations to all possible situations (Li et al., 2019). Additionally, COLREGS does not provide specific guidance for the actual operation of ships (Hilgert and Baldauf, 1997; Zhang et al., 2012; He et al., 2017; Du et al., 2020b) and does not define a well-clear distance for reducing the risk of collision (Hörteborn et al., 2019). Hence, collision assessment (CA) is a vital tool for the operational safety of ships, and many techniques have been developed (Zhang et al., 2017).

A collision occurs when two objects arrive at the same point in space and time. If the sizes of the two objects are not considered, the distance between the two objects when they collide is 0. However, ships are three-dimensional objects, and they differ in size with various lengths of 10–400 m. Collision risk assessments often use distance parameters between ships but rarely consider the dimensions of the ships themselves. The most commonly used distance parameter for CA is the distance to the closest point of approach (DCPA); this is frequently used together with the time to the closest point of approach (TCPA). The IMO recommends that ships install an automatic radar plotting aid (ARPA) (SOLAS, 1974), which can automatically calculate the DCPA and TCPA to a relative ship (RS) detected by radar. Navigators can then assess the collision risk for their own ship (OS) based on the DCPA and TCPA values. The International Convention on Standard of Training, Certification, and Watchkeeping for Seafarers requires navigators to be trained on how to use radar and an ARPA. However, the DCPA and TCPA have limited applicability to CA (Szlapczynski and Szlapczynska, 2017). For example, if an RS has a DCPA of 0 nautical miles and a TCPA of 1 h, this

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Nomenclature

Symbol	Definition
x_{os}, y_{os}	Coordinates of the OS in the Earth-fixed coordinate system
x^f_{os}, y^f_{os}	Coordinates of the OS forecastle point in the Earth-fixed coordinate system
x_{vso}, y_{vos}	Coordinates of the VOS in the Earth-fixed coordinate system
x^f_{vos}, y^f_{vos}	Coordinates of the VOS forecastle point in the Earth-fixed coordinate system
x_{rs}, y_{rs}	Coordinates of the RS in the Earth-fixed coordinate system
ψ_{os}, ψ_{rs}	Headings of the OS and RS
x_c, y_c	Coordinates of the collision point in the Earth-fixed coordinate system
RP	Rendezvous point
θ_{OR}	Intersection heading angle between ψ_{os} and ψ_{rs}
CDC	Computed Distance at Collision
OS	Own ship
RS	Relative ship
δ_{OS}	Rudder angle of the OS
T	Simulation time
L	Ship length
AD	Actual distance between the OS and RS
$\theta_{IC}\theta_{OR}$	at which collision is inevitable
β	Bearing angle between the OS and RS

situation is not considered a serious collision risk because the RS may change its course within this time. Similarly, if the TCPA is 0 h and the DCPA is 1 nautical mile, the collision risk is minimal because the RS is already 1 nautical mile away (Zheng et al., 2020).

Several researchers have developed various techniques to overcome the limitations of applying the DCPA and TCPA alone to CA. Hasegawa and Kouzuki (1987), Hasegawa (1989) presented a collision risk model that uses fuzzy logic when calculating the degree of collision risk with the DCPA and TCPA. Kijima and Furukawa (2001, 2002) used a fuzzy inference system to evaluate the collision risk with a non-dimensionalized DCPA and TCPA. Ahn et al. (2012) proposed a collision risk model that uses an adaptive network-based fuzzy inference system and multilayer perceptron. Perera et al. (2012) presented a collision risk model that comprises a decision-making module combining fuzzy logic and a Bayesian network. Wu et al. (2019) proposed a ship–bridge collision alert system that considers the ship parameters, bridge parameters, and natural environment. Park et al. (2011) utilized the Monte Carlo method and a multilayer perceptron neural network for CA. However, the fuzzy theory still has some limitations when applied to collision risk models (Li and Pang, 2013). Fuzzy algorithms use Boolean computation, so a ship with the greatest membership value for the collision risk is considered the most dangerous (Li and Pang, 2013). Zheng and Wu (2000) pointed out that unclear membership for the collision risk may lead to ambiguity in the risk assessment of RSs. Neural networks can have lengthy training times because of their complex structure (Li and Pang, 2013).

Fujii and Tanaka (1971) proposed the concept of a ship domain (i.e., the area around a ship) for CA. Various risk assessment methods that utilize the ship domain have since been developed (Goodwin, 1975; Davis et al., 1980; Coldwell, 1983; Zhao et al., 1993; Pietrzykowski, 2008; Liu et al., 2016; Szlapczynski and Szlzpcynska, 2017; Im and Luong, 2019).

The implementation of an automatic identification system (AIS) also allows efficient CA via AIS data. CA was studied by using AIS ship traffic data to identify ship collision or near-collision situations (Goerlandt and Kujala, 2011; Zhang et al., 2015, 2016; Gao and Shi, 2020). Even in a dangerous situation, it is now possible to analyze how a real ship behaves to avoid a collision (Mou et al., 2010). Furthermore, AIS data was used in a study of autonomous ship collision avoidance (Hörteborn and Ringsberg, 2021).

Park and Kim (2016) used the concept of probability flow for CA. Chai et al. (2017) developed a quantitative risk assessment model that considers all possible accident scenarios. Lenart (2015) proposed a new collision threat parameter (i.e., time to a safe distance) for application to the ARPA.

However, the above models for CA did not consider their application to the stand-on ship. Thus, they are difficult to apply to defining the timing when stand-on ships should take action to avoid a collision as stated in Rule 17 of the COLREGS (1972). Some studies have considered the safety of the stand-on ship (Szlapczynski et al., 2018; Du et al., 2020a, 2020b; Gil et al., 2020), but few have suggested specific numerical values for timing cooperative actions for collision avoidance. Szlapczynski et al. (2018) presented a study on collision avoidance of a stand-on ship based on the ship domain. Furthermore, Gil et al. (2020) used minimum distance to collision (MDTC) to propose a region where collisions cannot be avoided. The MDTC, on the other hand, was studied using ship domain, disk, and safe margin. And Du et al. (2020a, 2020b) studied to develop an alert system for a stand-on ship and improve the situational awareness of a stand-on ship in a dangerous situation based on protective layers such as ship domain.

This study will present a CA model between two ships that can consider all situations in which two ships collide directly without a safe margin, such as the ship domain. As a result, this study demonstrates a more detailed distance between the centers of two ships when they collide directly. The length and width of a ship, the angle of collision, and the collision point of the struck ship are carefully considered for this purpose.

A novel method for CA is needed to determine when the stand-on ship should take cooperative action. In this study, a criterion called the computed distance at collision (CDC), was developed. The CDC offers a specific numerical value that can be used to assess when a stand-on ship should take cooperative action for collision avoidance, and it can be used to define the minimum time for the stand-on ship to take action.

2. Methodology

This section will explain how to perform CA. CA requires the calculation of novel criteria, CDC. To that end, introduction of ship motion equations to obtain ship's trajectories, as well as how to set simulation scenarios, how to calculate the CDC, and the criterion required for CA, are presented first. The CA will also identify areas where collisions cannot be avoided.

2.1. Overview

Although several methods have been adopted for CA, the most common approach is based on the distance and time between two ships (Szlapczynski and Szlapczynska, 2017; Li et al., 2019). The distance can be used to assess the risk of a collision, and the time can be used to assess when to take avoidance actions in dangerous situations which is DCPA is smaller than safe distance and TCPA is smaller than safe time. For example, when two ships collide, the distance between them is close to 0. Thus, the collision risk can be determined by calculating the distance between the two ships. Meanwhile, if both ships arrive at the same position at the same time, they will collide. However, because ships are difficult to maneuver, a collision is inevitable unless an avoidance action is performed in time. Rule 8 of COLREGS recommends that actions to avoid a collision shall be made in ample time if circumstances permit. Hence, the collision risk can be assessed by calculating threshold values for both time and distance. To obtain these two parameters, the ships'

Table 1
Main parameters of the model ship.

Ship parameter	KVLCC2
Length, L (m)	2.5000
Breadth, B (m)	0.4531
Draft, d (m)	0.1625
Block coefficient, C_b	0.8098
Scale	128.0
Speed, U (m/s)	0.682

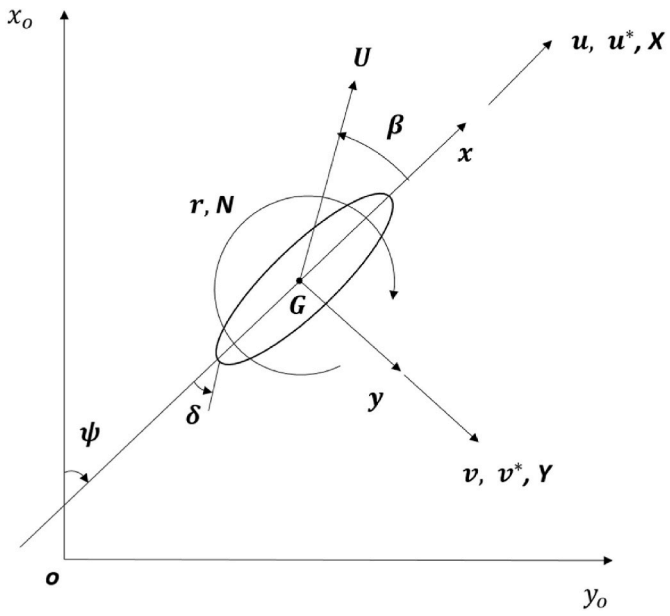


Fig. 1. Earth- and body-fixed coordinate systems.

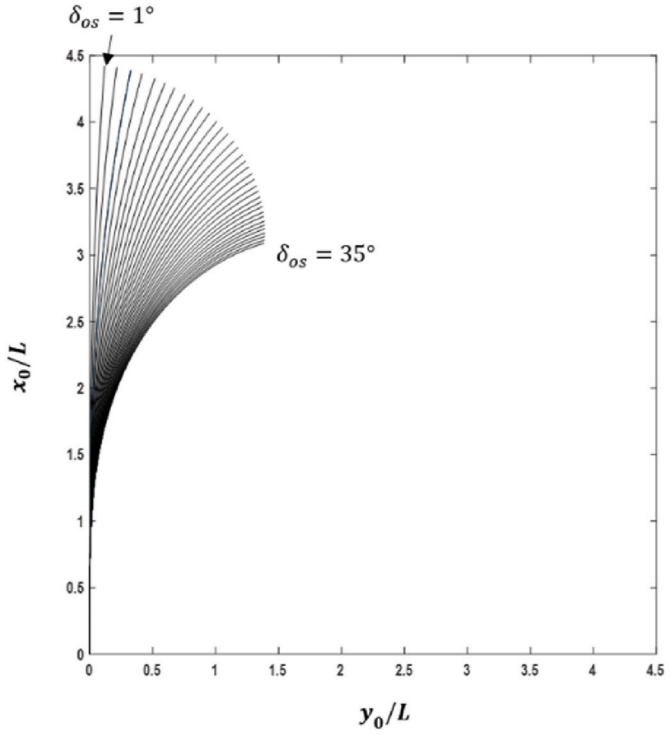


Fig. 2. Model ship trajectories.

coordinates are needed to calculate their distance, and various scenarios must be simulated.

In this study, the CDC was developed as a criterion for timing when a stand-on ship should perform a cooperative action for collision avoidance based on distance and time parameters. The reference point of a ship was defined as the midpoint along its length and breadth. The CDC is the distance between two ships at which collision becomes inevitable, and it is defined according to the ships' reference points. The CDC does not become 0 even if the two ships collide. Precise calculation of the CDC requires detailed information such as the position where the two ships contact, the relative angle between them, and differences in their length and breadth. In this study, the coordinates of the reference points were assumed to come from the Global Positioning System.

2.2. Equations of ship motion

The distance between two ships can be calculated according to their coordinates. The coordinates can be obtained from equations of ship motion based on the mathematical models of the hydrodynamic forces acting on a ship are required. In this study, the Maneuvering Modeling Group (MMG) model was adopted because of its accuracy and ability to consider the hydrodynamic forces acting on each element of a ship, such as a hull, propeller, and rudder (Kijima et al., 1990). Hydrodynamic coefficients to calculate hydrodynamic forces acting on the three elements are obtained through captive model tests. (Yoshimura, 2005). Zhang et al. (2017) noted that the MMG model offers superior performance to other models when considering the dynamics and actual conditions of a propeller and rudder, which is good for course-keeping control. Additionally, model ships were required to calculate the motions of ships during maneuvers. KVLCC2 was selected to model both the OS and RS, and its details are presented in Table 1.

Fig. 1 shows Earth- and body-fixed coordinate systems. Hydrodynamic forces acting on a ship are more effectively described in a body-fixed coordinate system (Yasukawa and Yoshimura, 2015). The equations of ship motions are given as follows

$$\left.\begin{aligned}
&\left(m' + m'_x\right)\left(\frac{L}{U}\right)\left(\frac{\dot{U}}{U}\cos\beta - \dot{\beta}\sin\beta\right) + \left(m' + m'_y\right)r'\sin\beta = X',\\
&-\left(m' + m'_y\right)\left(\frac{L}{U}\right)\left(\frac{\dot{U}}{U}\sin\beta - \dot{\beta}\cos\beta\right) + \left(m' + m'_x\right)r'\cos\beta = Y',\\
&\left(I'_{zz} + i'_{zz}\right)\left(\frac{L}{U}\right)^2\left(\frac{\dot{U}}{L}r' + \frac{U}{L}r'\right) = N',
\end{aligned}\right\} \quad (1)$$

where G is the center of gravity of the ship and X and Y are the forces along the x and y axes. N represents a moment around a vertical axis through G. u and v are the components of the ship speed U along the x and y axes, respectively. ψ is the heading angle, β is the drift angle, and δ is the rudder angle. The prime symbol ($'$) indicates a non-dimensionalized value, and a dot above () indicates time differentiation. m', m'_x, and m'_y are the mass and x- and y-axis components of the added mass of the ship. I'_{zz} and i'_{zz} are the moment and added moment of inertia of the ship, respectively. X', Y', and N' can then be expressed as follows:

$$\left.\begin{aligned}
X' &= X'_H + X'_P + X'_R,\\
Y' &= Y'_H + Y'_R,\\
N' &= N'_H + N'_R,
\end{aligned}\right\} \quad (2)$$

where H, P, and R represent the hull, propeller, and rudder components, respectively.

The mathematical model of non-dimensional hydrodynamic forces and the moment acting on the hull is given by (Kijima et al., 1990)

Table 2
Hydrodynamic derivatives and other coefficients of KVLCC2.

Item	Value	Item	Value
$X'_{\beta r}$	0.0898	N'_{rr}	0.0214
X'_{uu}	−0.0250	$N'_{\beta\beta r}$	−0.0228
Y'_{β}	0.2387	$N'_{\beta rr}$	−0.2660
$Y'_{\beta\beta}$	0.5928	m'_x	0.0220
$Y'_{\beta\beta r}$	0.2844	m'_y	0.2230
$Y'_{\beta rr}$	0.5367	t_R	0.2000
N'_{β}	0.1377	a_H	0.3120
N'_r	−0.0478	x'_H	−0.4640
$N'_{\beta\beta}$	−0.0533		

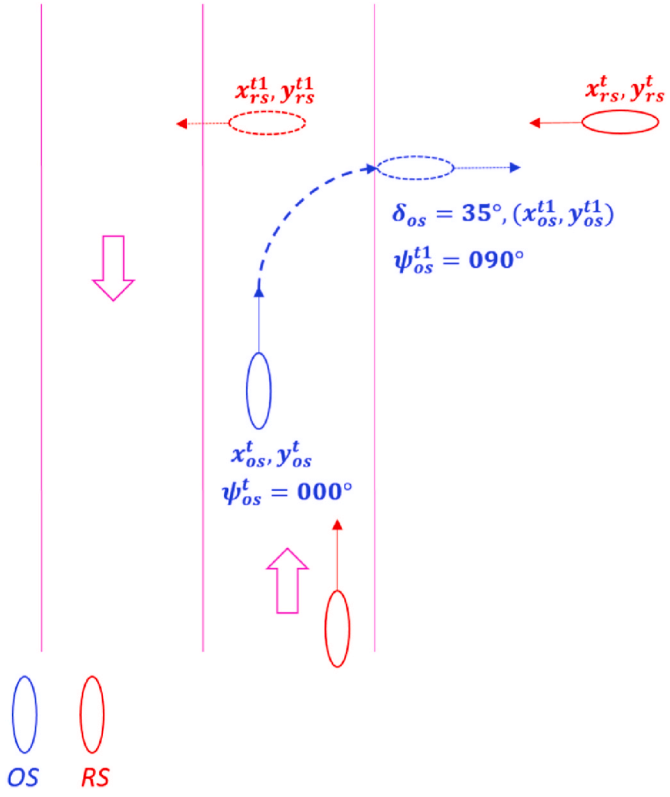


Fig. 3. Worst-case scenario.

$$\left.\begin{aligned} X'_H &= X'_{\beta r} r' \sin\beta + X'_{uu}\cos^2\beta, \\ Y'_H &= Y'_{\beta}\beta + Y'_r r' + Y'_{\beta\beta}\beta|\beta| + Y'_{rr} r'^{|r'|} + \left(Y'_{\beta\beta r}\beta + Y'_{\beta rr} r'\right)\beta r', \\ N'_H &= N'_{\beta}\beta + N'_r r' + N'_{\beta\beta}\beta|\beta| + N'_{rr} r'^{|r'|} + \left(N'_{\beta\beta r}\beta + N'_{\beta rr} r'\right)\beta r', \end{aligned}\right\} \quad (3)$$

$X'_{\beta r}$ is the variation in the longitudinal force according to the drift angle β and non-dimensional yaw rate r'; X'_{uu} is the resistance of the ship to forward straight motion; and Y'_{β}, Y'_r, N'_{β}, and N'_r are linear hydrodynamic derivatives.

The longitudinal force produced by a propeller X'_P is given by

$$\left.\begin{aligned} X'_P &= (1-t_P)n^2 D_P^4 K_T(J_P) \Big/ \frac{1}{2} L d U^2, \\ K_T(J_P) &= C_1 + C_2 J_P + C_3 J_P^2, \\ J_P &= U\cos\beta(1-w_P)/nD_P, \end{aligned}\right\} \quad (4)$$

where t_P is the thrust deduction coefficient; n is the number of propeller revolutions; D_P is the diameter of the propeller; $K_T(J_P)$ is the thrust coefficient; J_P is the advance coefficient; $C_1, C_2,$ and C_3 are constants; and w_P is the effective wake fraction at the propeller position.

The hydrodynamic forces and moment generated by the rudder are given by

$$\left.\begin{aligned} X'_R &= -(1-t_R)F'_N \sin\delta, \\ Y'_R &= -(1+a_H)F'_N \cos\delta, \\ N'_R &= -\left(x'_R + a_H x'_H\right)F'_N \cos\delta, \end{aligned}\right\} \quad (5)$$

where F'_N is the non-dimensional rudder normal force, t_R and a_H are interaction coefficients, and x'_R and x'_H are respectively the positions of the rudder and the acting point of the additional lateral force induced by steering.

Table 2 shows the hydrodynamic derivatives and other coefficients used in simulations are obtained from captive model tests of KVLCC2 carried out at the Seakeeping and Maneuvering Basin in Kyushu University as shown in Table 2. The accuracy of the simulations of maneuvering motions using the MMG model was confirmed by comparing simulated turning trajectories with measured turning trajectories obtained in advance by conducting free-running model tests in the basin.

The above equations can be used to calculate the trajectory of a model ship based on simultaneously solving the differential equations in Eq. 1. Fig. 2 shows the trajectories of the KVLCC2 model ship in the Earth-fixed coordinate system when the sailing time was fixed and the rudder angle was increased from 1.0° to 35.0° in increments of 1°. The trajectories are expressed as non-dimensional values on the basis of dividing the distance by the ship length.

2.3. Simulations

The coordinates of the OS and RS can be used to simulate various collision scenarios. The initial positions of the OS and RS were set so that a collision would occur if both ships took no action. The OS was set as the give-way ship, and the RS was set as the stand-on ship. The trajectories of the give-way ship were obtained by solving the equations of ship motion with several rudder angles to represent collision avoidance actions. By contrast, the stand-on ship was set to maintain its speed and course. To avoid a collision between the two ships, the give-way ship should use a rudder. However, there are situations where a collision between the two ships is inevitable. If those situations can be identified and the initial states of the two ships are known, the area where inevitable collisions occur can be defined. To obtain the inevitable collision area, the collision between the two ships was assessed by using the CDC and actual distance (AD). The CDC was calculated by moving the OS from its original position to a virtual position where it collides with the RS. When the OS is moved to the virtual position, its heading angle should not be changed.

The simulation scenarios were generated according to the following constraints:

1. The OS was the give-way ship and the RS was the stand-on ship.
2. The sailing area in restricted areas was spatially set to be limited due to the heavy traffic volume, the depth, and the islands around the restricted area.
3. Among the three types of encounters (i.e., crossing, head-on, and overtaking), only the crossing encounter was considered. In a head-on encounter, the OS should be able to take actions that would avoid a collision. The overtaking encounter was also excluded because the two ships had the same speed. Thus, various crossing encounters were considered by varying the angle between the headings of the two from 3.0° to 112.5° (Cockcroft and Lameijer, 2003).
4. Considering that the ships were operating in a restricted area, the OS was assumed to be able to change its heading angle by a maximum of

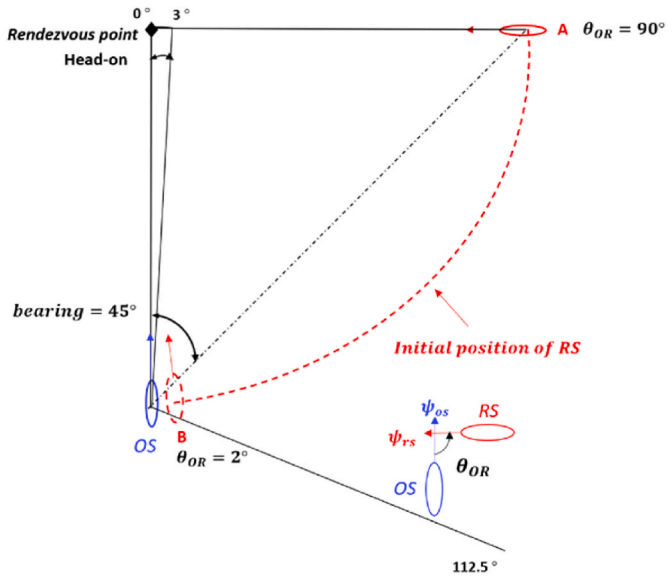


Fig. 4. Range of encounters considered in the simulation scenarios.

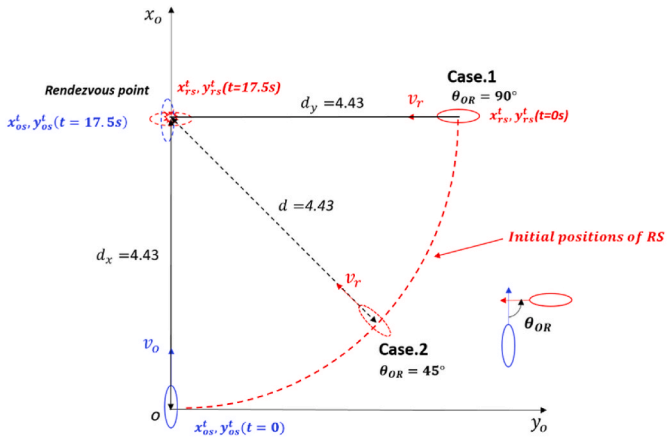


Fig. 5. Setting the distances of the two ships from the RP.

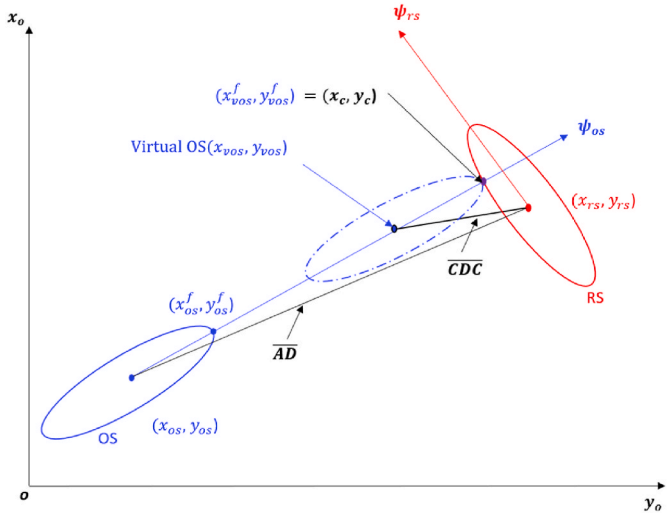


Fig. 6. Example calculation of the CDC.

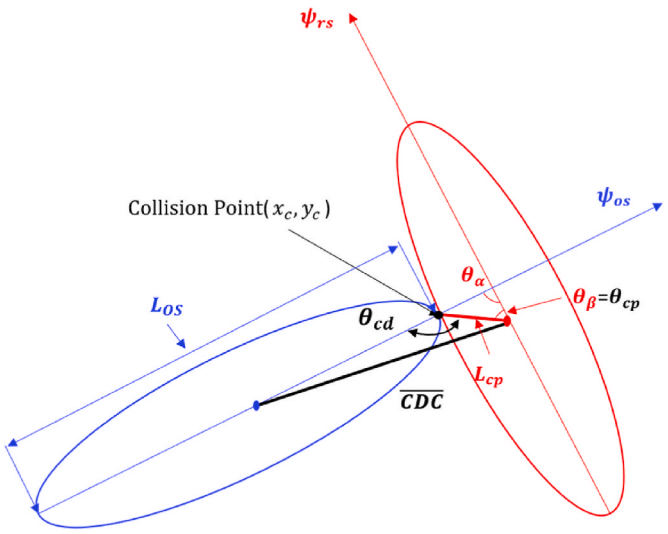


Fig. 7. Parameters used for CDC calculation.

90.0° to avoid the RS. Changing the heading angle by more would cause the OS to navigate backward, which would raise the possibility of a collision with ships approaching from behind.

Fig. 3 shows the worst-case scenario. x_{os}^t and y_{os}^t are the coordinates of the OS position at time t, and x_{rs}^t and y_{rs}^t are the coordinates of the RS position at time t. ψ_{os}^t is the heading angle of the OS at time t, and δ_{os} is the rudder angle of the OS. The OS was assumed to vary the rudder angle from 0.0° to 35.0°, and it took 17.5 s for the OS to change its heading by 90.0° with the maximum rudder angle. Therefore, the simulation period was fixed to 17.5 s with intervals of 0.5 s. Simulations were executed assuming no external disturbances such as current or wind.

Fig. 4 shows the encounters considered in the simulation scenarios. The position at which the two ships meet was defined as the rendezvous point (RP). The two ships met at the RP after 17.5 s. As noted earlier, only crossing encounters were considered. The red dotted line indicates the candidate initial positions for the RS. θ_{OR} is the intersection heading angle between the headings of the OS (ψ_{os}) and RS (ψ_{rs}). θ_{OR} was increased in intervals of 2.0° from a minimum of 2.0° to a maximum of 90.0°. Point A is the position of the RS when $\theta_{OR} = 90.0°$, and Point B is when $\theta_{OR} = 2.0°$. The heading of the RS was set to face the RP.

Fig. 5 shows the initial positions of the OS and the RS. The initial position of the OS was set to the origin O of the Earth-fixed coordinate system. The distance between O and the RP was set to 4.43 L, and the red dotted line shows an arc with a radius of 4.43 L around the RP. The initial position of the RS was selected along the red dotted line by changing θ_{OR} between 2.0° and 90.0°. For example, Case 1 represents the RS position when $\theta_{OR} = 90.0°$, and Case 2 represents the RS position when $\theta_{OR} = 45.0°$.

2.4. Calculations of the CDC and AD

2.4.1. General

The simulation scenarios were used to calculate distances between the OS and RS such as the AD and CDC. The AD between the OS and RS was obtained by calculating the coordinates of the reference points between the two ships. All calculations related to the distance between the two ships were done by using the Earth-fixed coordinate system shown in Fig. 1. To calculate the CDC, the OS and RS were assumed to collide with each other. A virtual own ship (VOS) with the same heading angle as the OS was introduced at the position where its forecastle touches the

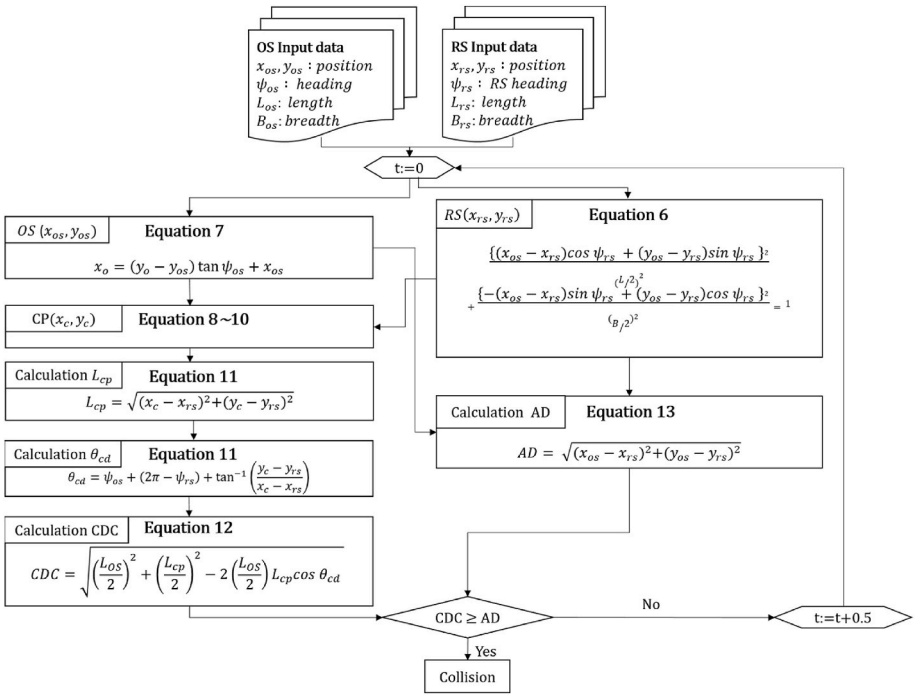


Fig. 8. CA algorithm.

RS. This position was defined as the collision point (CP).

Fig. 6 shows an example calculation of the CDC. x_{os} and y_{os} are the coordinates of the reference point of the OS, and x^f_{os} and y^f_{os} are the coordinates of the OS forecastle, which is at the front end of its centerline. The coordinates of the reference point of the VOS are indicated by x_{vos} and y_{vos}, and x^f_{vos} and y^f_{vos} are the coordinates of the VOS forecastle. The coordinates of the reference point of the RS are x_{rs} and y_{rs}. The CP coordinates are represented by x_c and y_c in the Earth-fixed coordinate system. ψ_{os} and ψ_{rs} are the headings of the OS and RS, respectively. Both ships were represented by ellipses for simplicity. The lengths of the major and minor axes of the ellipses corresponded to the ship's length and breadth, respectively. To calculate the CDC, the centerline of the OS was extended in the Earth-fixed coordinate system. The CP coordinates x_c and y_c can be defined as the intersection between the extended centerline and the elliptical shape of the RS. The two ships do not collide when the AD is greater than the CDC because the OS has room to avoid the RS. Thus, the CDC can be the criterion for CA.

2.4.2. Procedure for calculating the CDC and AD

The equations of an ellipse representing the RS at (x_{rs}, y_{rs}) with the heading angle ψ_{rs} and the extended centerline of the OS at (x_{os}, y_{os}) with the heading angle ψ_{os} are expressed in the Earth-fixed coordinate system as follows:

$$\frac{\{(x_o - x_{rs})\cos\psi_{rs} + (y_o - y_{rs})\sin\psi_{rs}\}^2}{(L/2)^2} + \frac{\{-(x_o - x_{rs})\sin\psi_{rs} + (y_o - y_{rs})\cos\psi_{rs}\}^2}{(B/2)^2} = 1. \tag{6}$$

$$x_o = (y_o - y_{os})\tan\psi_{os} + x_{os}. \tag{7}$$

Eq. (6) depicts the equation of an ellipse representing the RS, while Eq. (7) depicts the equation of the OS extended centerline in the Earth-fixed coordinate systems $o - x_0y_0$. If the extended centerline intersects the ellipse, the right-hand side of Eq. (7) can be substituted into x_o in Eq. (6) to obtain the coordinates of the CP. If the extended centerline does not intersect the ellipse, the condition does not have the CDC.

Substituting Eq. (7) into Eq. (6), a quadratic equation can be obtained as shown in Eq. (8).

$$Ay_o^2 + 2By_o + C = 0 \tag{8}$$

Where, A, B and C are given as follows.

$$\left.\begin{aligned}
A &= \left[\left\{\left(\cos^2\psi_{rs}\right) + \left(2\tan\psi_{rs}\cos\psi_{rs}\sin\psi_{rs}\right) + \left(\tan^2\psi_{rs}\sin^2\psi_{rs}\right)\right\}(B/2)^2\right. \\
&\left. + \left\{\left(\tan^2\psi_{rs}\cos^2\psi_{rs}\right) - \left(2\tan\psi_{rs}\cos\psi_{rs}\sin\psi_{rs}\right) + \left(\sin^2\psi_{rs}\right)\right\}(L/2)^2\right], \\
B &= \left[\left\{-\left(2y_{rs}\cos^2\psi_{rs}\right) + \left(2x_{os}\cos\psi_{rs}\sin\psi_{rs}\right) - \left(2x_{rs}\cos\psi_{rs}\sin\psi_{rs}\right)\right.\right. \\
&\left. - \left(2x_{os}y_{rs}\cos\psi_{rs}\sin\psi_{rs}\right) + \left(2\tan\psi_{rs}q\sin^2\psi_{rs}\right) - \left(2x_{os}y_{rs}\sin^2\psi_{rs}\right)\right\}(B/2)^2 \\
&+ \left\{\left(2\tan\psi_{rs}x_{os}\cos^2\psi_{rs}\right) - \left(2\tan\psi_{rs}x_{rs}\cos^2\psi_{rs}\right) - \left(2x_{os}\cos\psi_{rs}\sin\psi_{rs}\right)\right. \\
&\left.\left. + \left(2x_{rs}\cos\psi_{rs}\sin\psi_{rs}\right) + \left(2\tan\psi_{rs}y_{rs}\cos\psi_{rs}\sin\psi_{rs}\right) - \left(2y_{rs}\sin^2\psi_{rs}\right)\right\}(L/2)^2\right], \\
C &= \left[\left\{\left(y_{rs}^2\cos^2\psi_{rs}\right) - \left(2x_{os}y_{rs}\cos\psi_{rs}\sin\psi_{rs}\right) + \left(2x_{rs}y_{rs}\cos\psi_{rs}\sin\psi_{rs}\right)\right.\right. \\
&\left. + \left(x_{os}^2\sin^2\psi_{rs}\right) - \left(2x_{rs}x_{os}\sin^2\psi_{rs}\right) + \left(x_{rs}^2\sin^2\psi_{rs}\right)\right\}(B/2)^2 + \left\{\left(x_{os}^2\cos^2\psi_{rs}\right)\right. \\
&\left. - \left(2x_{os}x_{rs}\cos^2\psi_{rs}\right) + \left(x_{rs}\cos^2\psi_{rs}\right) + \left(2x_{os}y_{rs}\cos\psi_{rs}\sin\psi_{rs}\right)\right. \\
&\left.\left. - \left(2x_{rs}y_{rs}\cos\psi_{rs}\sin\psi_{rs}\right) + \left(y_{rs}^2\sin^2\psi_{rs}\right) - \left((B/2)^2\right)\right\}(L/2)^2\right]
\end{aligned}\right\} \tag{9}$$

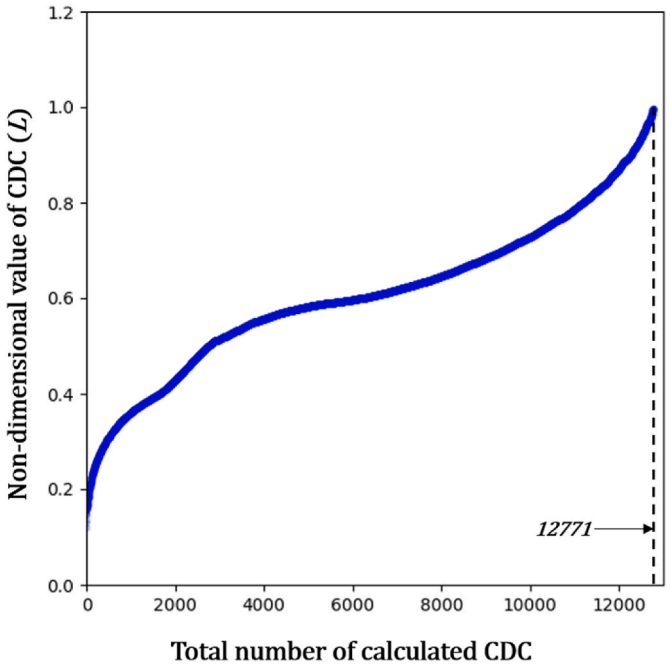


Fig. 9. CDC distribution.

Finally, y_0 is given by

$$y_o = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \quad (y_c \in y_o). \tag{10}$$

Eq. (10) yields two y_c values. When the extended centerline of the OS and the equation of an ellipse representing the RS come into contact, two intersection points exist, as shown in Fig. 6. One point is on the RS's port side, and the other is on its starboard side. Because the OS is coming from the port side of the RS in the scenario depicted in Fig. 6, collisions occur on the port side of the RS. The Earth-fixed coordinate system is used to calculate the CP coordinates x_c and y_c. As a result, a lower value of y_c indicates a CP on the port side of the RS in this study.

Once the CP coordinates are obtained, the CDC can be calculated. Fig. 7 presents the parameters necessary for CDC calculation. L_{OS} is the length of the OS, and L_{cp} is the distance between the RS coordinates and CP coordinates. θ_{cd} is the angle between the extended centerline of the OS and a line connecting the CP and the reference point of the RS. L_{cp} and θ_{cd} can be calculated as follows:

$$\left.\begin{aligned} L_{cp} &= \sqrt{(x_c - x_{rs})^2 + (y_c - y_{rs})^2}, \\ \theta_{cd} &= \psi_{os} + (2\pi - \psi_{rs}) + \tan^{-1}\left(\frac{y_c - y_{rs}}{x_c - x_{rs}}\right). \end{aligned}\right\} \tag{11}$$

Then, the CDC can be calculated as follows:

$$CDC = \sqrt{\left(\frac{L_{OS}}{2}\right)^2 + (L_{cp})^2 - 2\left(\frac{L_{OS}}{2}\right)L_{cp}\cos\theta_{cd}}\,. \tag{12}$$

The AD between two ships can be calculated by using the coordinates of the two ships:

$$AD = \sqrt{(x_{os} - x_{rs})^2 + (y_{os} - y_{rs})^2}. \tag{13}$$

2.5. Summary

Once coordinates and headings of the two ships are known, the CDC can be obtained. Then, the collision can be assessed by comparing the AD and CDC between the two ships. Fig. 8 shows the algorithm of the whole calculation process of CA.

The CDC is calculated through the following process. At the moment of a collision, as shown in Fig. 7, a triangle is formed using L_{OS}, L_{cp} and a line segment connecting the two ships. L_{cp} and θ_{cd} are calculated using the parameters of the two ships as shown in Eq. (11). The CDC was then calculated as the length of the line segment between the ships using Eq. (12).

The AD can be easily calculated using the coordinates of the two ships. So, if the AD is less than or equal to the CDC, the two ships are expected to collide.

3. Results

CA is performed in this section by comparing the CDC and AD values. Simulations of two ships' motion are performed by the scenarios depicted in Fig. 5. As shown in Fig. 2, the OS increases the rudder angle by 1°, from 1° to 35°, while the RS maintains her speed and course, which are the duty of a stand-on ship, and only the initial position is altered. As a result, whether there is an initial position of the RS that cannot avoid collisions between the two ships will be described. And, if an unavoidable area exists, it will be described whether the timing of cooperation avoidance of stand-on ship can be suggested. Furthermore, dependability of the CDC will be verified.

3.1. CDC

The CDC was calculated for 58,320 scenarios of different combinations of δ_{os} (36 values), T (36 values), and θ_{OR} (45 values). Fig. 9 shows the CDC distribution for all scenarios. The lateral axis shows the accumulated number of CDCs. The CDC could only be calculated for situations where the extended centerline of the OS contacted the ellipse representing the RS, as shown in Fig. 6. Hence, the CDC was obtained for 12,771 scenarios. The vertical axis shows the CDC, which was non-dimensionalized by the ship length L. The maximum CDC was approximately 0.998 L, and this did not exceed L; the minimum CDC was approximately 0.120 L.

3.2. AD

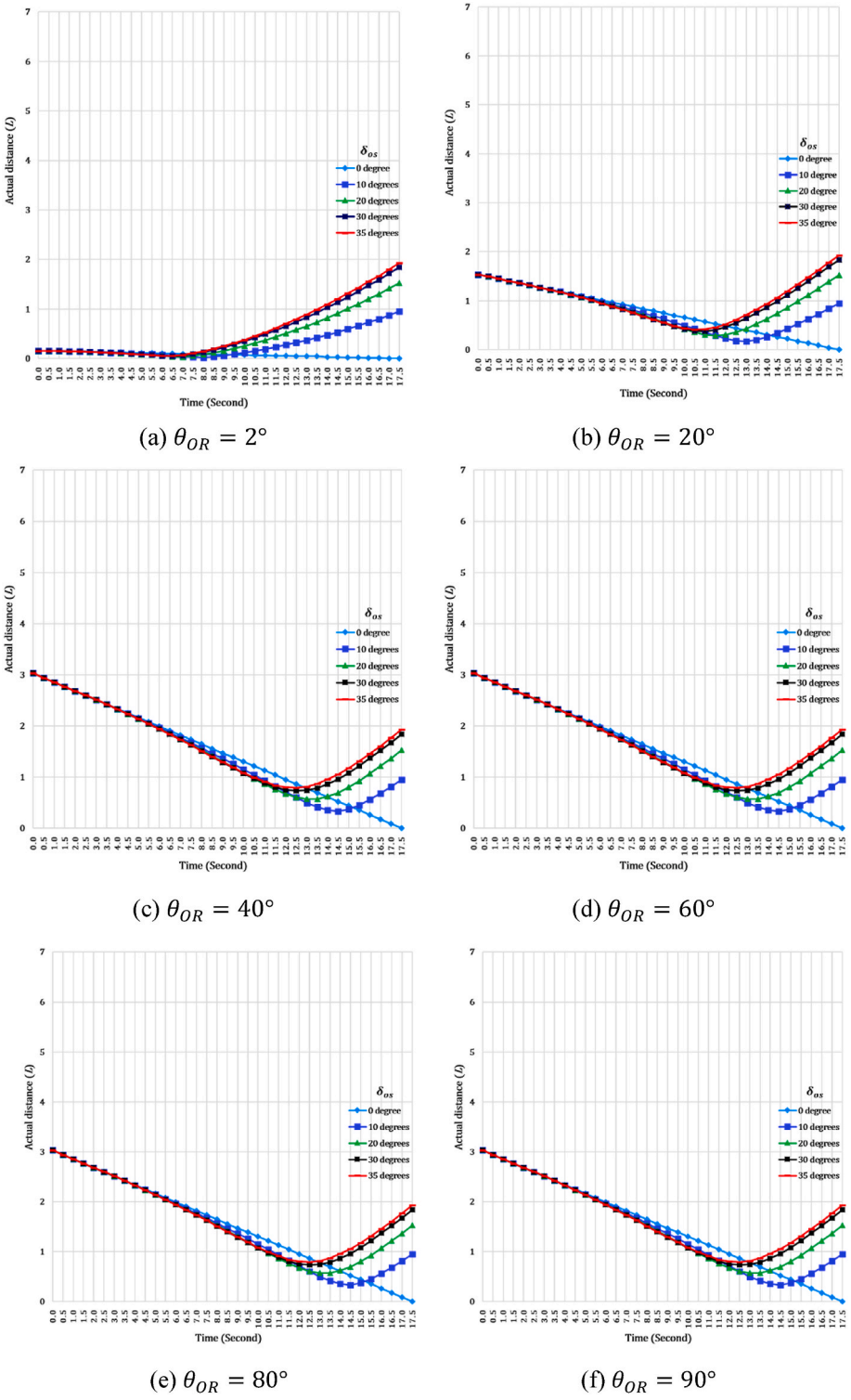
Fig. 10 shows the AD at different rudder angles for the OS from 0.0° to 35.0° at each θ_{OR}. The lateral axis indicates the elapsed simulation time, and the vertical axis shows the non-dimensional AD at each time. When the OS did not use the rudder (i.e., a rudder angle of 0°), the AD became 0L at 17.5 s. By contrast, when the OS did use the rudder, the two ships tended to approach each other and then separate. As shown in Fig. 9, the maximum CDC was less than 1L, so there is a collision risk if the AD is less than 1L.

3.3. CA results

As shown in Fig. 6, if the coordinates of the two ships at the same moment are known, the AD and CDC can be calculated for that moment. Hence, the collision risk can be predicted by comparing the AD and CDC.

Fig. 11 presents the CA results in (a) three-dimensional form and (b) two-dimensional form. The horizontal axes show the values of θ_{OR} and δ_{os}, and the vertical axis represents the CA value for each combination of θ_{OR} and δ_{os}. The CA value is 1 for collision cases and 0 for cases with no collision. These cases are colored red and blue, respectively. When CA = 1 at an arbitrary θ_{OR}, then the OS cannot avoid a collision with the RS regardless of the rudder angle δ_{os}. Therefore, Fig. 11(a) shows the approximate range where collisions are inevitable for arbitrary θ_{OR}. Fig. 11(b) shows the inevitable collision range more clearly, which appears to be θ_{OR} = 2.0°–34.0°. To apply this range of θ_{OR} to COLREGS, it must be converted into the bearings of the two ships.

Fig. 12 shows the conversion of the inevitable collision angle (θ_{IC}) to the bearing between two ships for COLREGS application. In crossing encounters, θ_{IC} converts to a bearing angle of 73.0°–89.0°. This defines

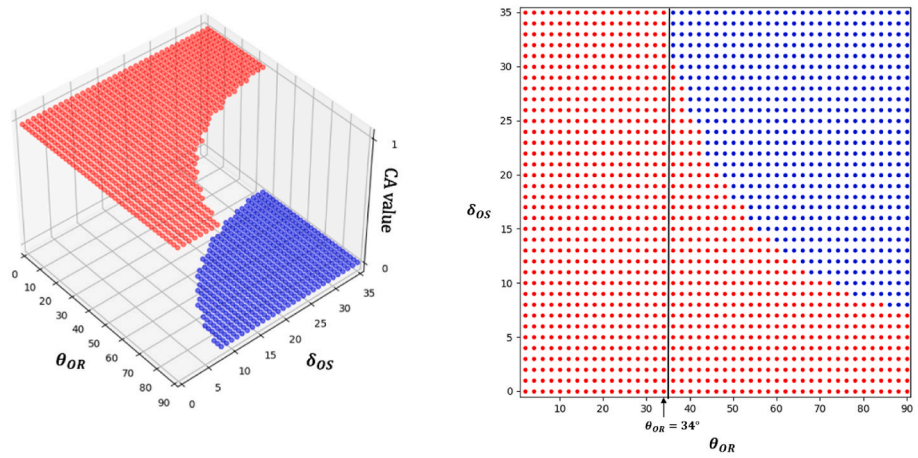


(a) $\theta_{OR} = 2°$ (b) $\theta_{OR} = 20°$

(c) $\theta_{OR} = 40°$ (d) $\theta_{OR} = 60°$

(e) $\theta_{OR} = 80°$ (f) $\theta_{OR} = 90°$

Fig. 10. AD between the OS and RS



(a) Three-dimensional expression (b) Two-dimensional expression

Fig. 11. CA results.

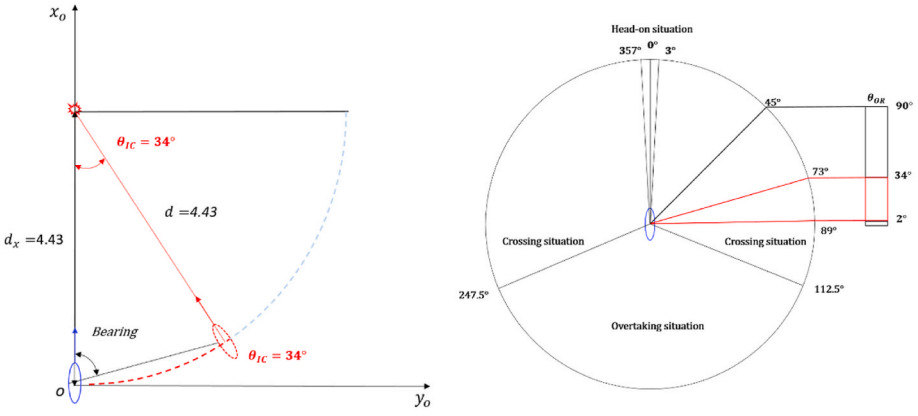
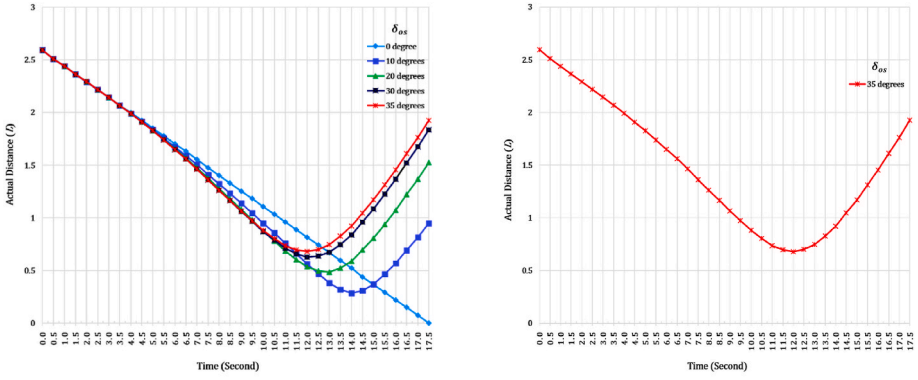
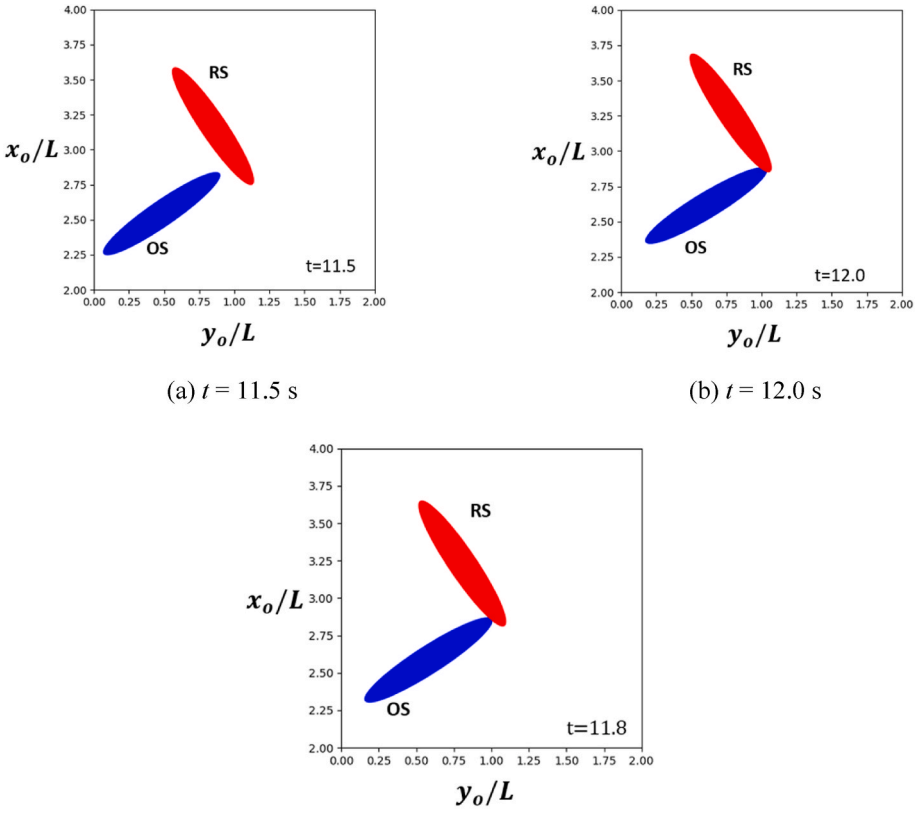


Fig. 12. Inevitable collision area converted for COLREGS application.



(a) AD at each rudder angle (b) AD when $\delta_{os} = 35°$

Fig. 13. AD between two ships at.$\theta_{OR} = 34°$



(a) t = 11.5 s (b) t = 12.0 s

(c) Moment of collision.

Fig. 14. Scenario where $\theta_{OR} = 34^\circ$ and.$\delta_{os} = 35^\circ$

Table 3
CDC, AD, and CA results for the scenario in Fig. 14

Time (s)	CDC (L)	AD (L)	CA result
11.5	0.6653	0.6975	No collision
11.8	0.6842	0.6830	Collision
12.0	0.6920	0.6787	Collision

the area where collision is inevitable between two ships in the given situation and time.

3.4. Analysis of the CDC

Collisions occurred in several of the simulation scenarios, and the CA results shown in Fig. 11 confirmed that $\theta_{IC} = 2.0^\circ$–34.0°. When $\theta_{OR} > 36.0^\circ$, collisions did not occur when $\delta_{os} > 31.0^\circ$, as shown in Fig. 11(b). The collision scenarios were then analyzed in more detail to obtain further insights.

Fig. 13(a) shows the AD at $\theta_{OR} = 34.0^\circ$, and Fig. 13(b) shows the AD at $\delta_{os} = 35.0^\circ$. The horizontal axis shows the elapsed time, and the vertical axis shows the AD between the two ships. The AD was 0L when $\delta_{os} = 0.0^\circ$ at $t = 17.5$ s. By contrast, the AD was approximately 2.0L when $\delta_{os} = 35.0^\circ$ at $t = 17.5$ s. When the OS had a rudder angle of $\delta_{os} = 35.0^\circ$, the minimum AD was approximately 0.6L at $t = 12.0$ s. This confirmed the CA results at $t = 12.0$ s, as shown in Fig. 14.

Fig. 14 shows the positions of the OS and RS before, after, and moment of a collision when $\theta_{IC} = 34.0^\circ$ and $\delta_{OS} = 35.0^\circ$. Ellipses which show simplified ship shapes are drawn in the same scale of vertical and lateral axes. Namely, the length of the ellipse is 1.0 and its width is 0.1812 which is equal to B/L. The two ships had not collided yet at 11.5 s but collided at 12.0 s. The exact collision time is 11.8 s as shown in Fig. 14(c). Table 3 presents the CDC and AD before, after, the moment of collision, and the CA results. The CDC was greater than the AD at 11.8 s, so a collision occurred at this time.

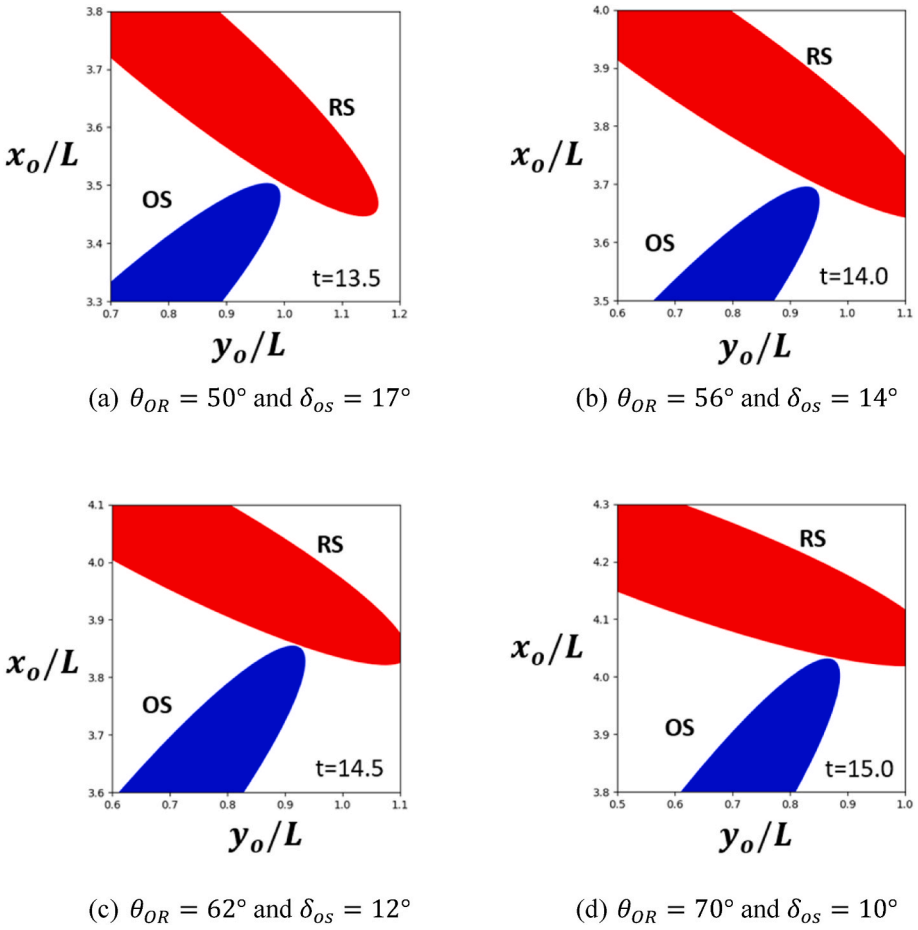
The CA was conducted by using the CDC. However, the reliability of the CDC must be further verified for several scenarios. Hence, the CDC and AD were compared for various scenarios in which the two ships would come very close to each other.

Fig. 15 shows cases of close encounters between two ships. The blue ellipses represent the positions of the OS, and the red ellipses represent the positions of the RS. Fig. 16 shows the difference between the values of AD and CDC. The horizontal axis is time and the vertical axis is the difference between AD and CDC and its unit is ship length L. Table 4 presents the CDC, AD, and CA results. No collisions occurred in all cases although the two ships were very close to each other. In Table 4, all ADs were greater than the corresponding CDCs by 0.004–0.013 L, which indicated that "no collision." would occur.

If the value of the vertical axis in Fig. 16 is 0 or less, a collision between the two ships has occurred. The value of the vertical axis in Fig. 16 is greater than 0 at the time shown in Table 4. The collisions in the four cases depicted in Fig. 16 occurred after the time specified in Table 4. This demonstrates the CDC's dependability.

4. Discussion

In this study, a novel CA method was developed that is based on using the CDC as an evaluation criterion. This made it possible to



(a) $\theta_{OR} = 50°$ and $\delta_{os} = 17°$ (b) $\theta_{OR} = 56°$ and $\delta_{os} = 14°$

(c) $\theta_{OR} = 62°$ and $\delta_{os} = 12°$ (d) $\theta_{OR} = 70°$ and $\delta_{os} = 10°$

Fig. 15. Cases of close encounters between two ships.

Table 4
CDC, AD, and CA results for the cases shown in Fig. 15

Case	θ_{OR}	δ_{os}	Time (s)	CDC (L)	AD (L)	AD - CDC (L)	CA
(a)	50°	17°	13.5	0.6382	0. 6513	0.0131	No collision
(b)	56°	14°	14.0	0.6197	0.6266	0.0069	No collision
(c)	62°	12°	14.5	0.6065	0.6124	0.0059	No collision
(d)	70°	10°	15.0	0.5867	0.5910	0.0043	No collision

accurately assess collisions between two ships even when they are in very close proximity that cannot be identified by ship domains or CPA. The results confirmed the reliability of the CDC, but its application may have limitations. Although it did not have a significant effect on θ_{IC}, the CDC was demonstrated to be inapplicable for the case where the extended centerline of the OS does not meet the ellipse represented by the RS but a collision occurs. The extended centerline passes through only the center of the OS, so it is impossible to identify whether the port side of the bow of the OS touches the stern of the RS.

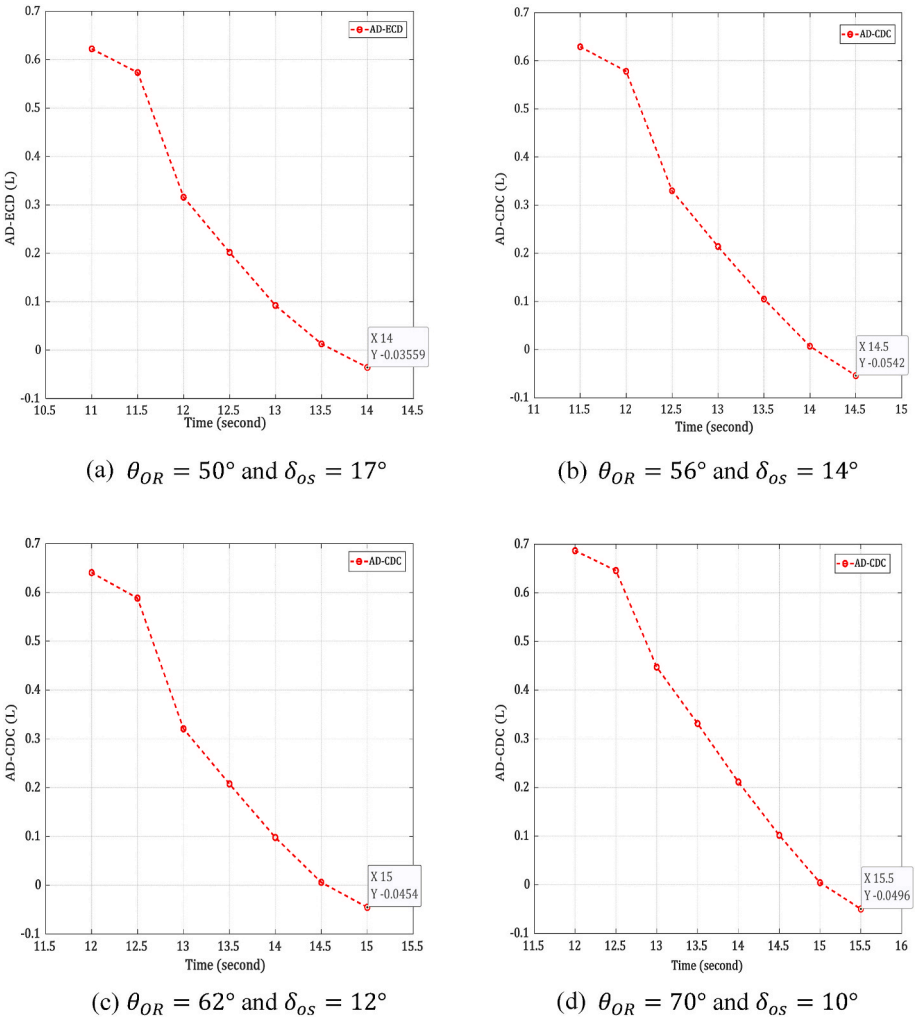
Fig. 17 shows cases in which the two ships collided, and Table 5 presents the relevant values when the collision occurred. At the moment of collision, the collision could be properly assessed as shown in Table 5. But, the extended centerline of the OS did not meet the ellipse representing the RS, so the CDC could not be calculated. These cases involved the OS brushing through the stern of the RS. This can be addressed by extending the centerlines of the port and starboard sides of the OS or, conversely, by using the extended centerline of the RS and the ellipse representing the OS. Furthermore, if the time interval of collision assessment is adjusted by 0.1 s, the limitation of CDC can be overcome.

We obtained trajectories of KVLCC2 ship models through equations of ship motion. The trajectories were then used to calculate the CDC based on the distances at which collision would inevitably occur depending on the angles of the two ships. The CDC was then used to calculate the inevitable collision area for the OS and RS. Fig. 18 visualizes the inevitable collision area with a solid red solid. If the distance between the two ships is within 2.59L and θ_{IC} is within 34.0°, two KVLCC2s sailing at the same speed cannot avoid a collision only by the action of the give-way ship within a period of 17.5 s.

However, there are several things to be further considered in this study, because the ships' trajectories can be changed by hull-hull interaction when two ships are in close proximity, or external disturbances such as wind, current, and so on in a restricted area.

If two ships approach each other close, hydrodynamic interaction occurs between their hulls which may affect their maneuvering motions. The degree of its influence on the maneuvering motions depends on relative positions and speed of the two ships. Hull-hull interaction mainly occurs when the two ships pass on lateral sides. However, in this study, situations such as shown in Figs. 14, 15 and 17 are considered to assess the possibility of collisions. At such situations, influence of hull-hull interaction is generally small.

On the other hand, if the external disturbances are reflected on ship's motion, the results can be changed. Therefore, the influence of external



(a) $\theta_{OR} = 50°$ and $\delta_{os} = 17°$ (b) $\theta_{OR} = 56°$ and $\delta_{os} = 14°$

(c) $\theta_{OR} = 62°$ and $\delta_{os} = 12°$ (d) $\theta_{OR} = 70°$ and $\delta_{os} = 10°$

Fig. 16. Gap of between AD and CDC

disturbances on the novel CA method should be investigated in our future work. By establishing scenarios based on what should be additionally considered, it will be possible to present a more sophisticated avoidance action timing of the stand-on ship in the restricted area in the future through the CDC model.

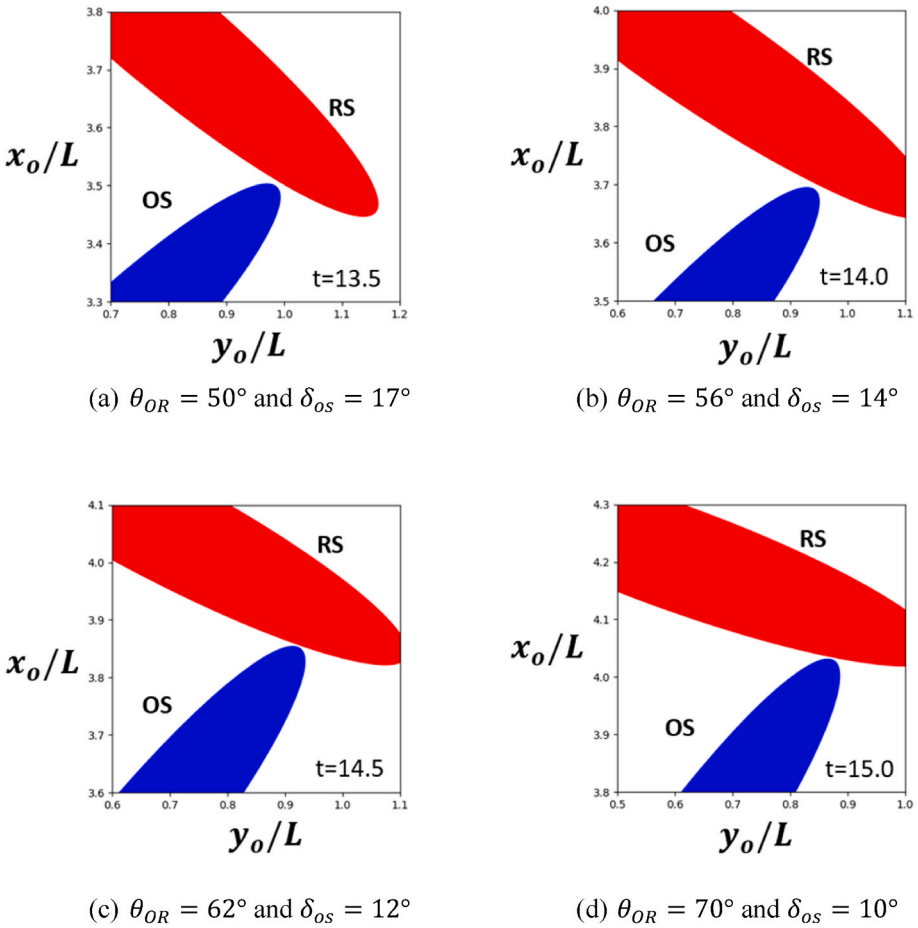
5. Conclusions

As Rule 17(b) of the COLREGS states, "When, from any cause, the vessel required to keep her course and speed finds herself so close that collision cannot be avoided by the action of the give-way vessel alone, she shall take such action as will best aid to avoid a collision." The stand-on ship must maintain its course and speed until the risk of collision is imminent. The duty of cooperative operation arises when the risk of collision is imminent (Cockcroft and Lameijer, 2003). However, this rule does not propose a specific timing for determining when the risk of collision is imminent.

As a result, the goal of this study was to present specific values for when the stand-on ship should take action to avoid a collision. This study computed the distance at collision between two ships to present the timing. The CDC was created using the intersection heading angle of the two ships, the RS's CP, the length of the OS, and the width of the RS. The CDC can evaluate ship collisions. Furthermore, simulations were performed to see if there were any areas where collisions were unavoidable based on the CDC. The simulations revealed that a collision could not be avoided purely by the action of the give-way ship when two ships sailed toward the same point at the same speed and with θ_{OR} less than 34°. Finally, it was concluded that if the value of θ_{OR} is less than 34°, the stand-on ship must take cooperative collision avoidance actions.

The results of this study are limited to the scenarios based on the same model ships under calm water conditions considered in this research. Additionally, the hull-to-hull interactions between the two ships and hull-propeller-rudder-engine interactions of ships were not taken into account, and the MMG model may have a little error.

However, this study is meaningful because it applies an engineering approach to the COLREGS and provides numerical values for timing when the stand-on ship should take cooperative action. These numerical values could help judge the timing for a manned or unmanned stand-on ship to avoid a collision. One limitation of the proposed approach is when the extended centerline of the OS does not contact the RS but they still collide (e.g., the two ships brush each other), as mentioned earlier. Although this did not have a significant impact on determining the inevitable collision angle, this limitation should be addressed in future research to further increase the accuracy of the proposed approach.



(a) $\theta_{OR} = 50°$ and $\delta_{os} = 17°$ (b) $\theta_{OR} = 56°$ and $\delta_{os} = 14°$

(c) $\theta_{OR} = 62°$ and $\delta_{os} = 12°$ (d) $\theta_{OR} = 70°$ and $\delta_{os} = 10°$

Fig. 17. Case of limitation of CDC.

Table 5
CDC, AD, and CA results for cases (a) and (b) in Fig. 17

Case	θ_{OR}	δ_{os}	Time (s)	CDC (L)	AD (L)	Assessment result
(a)	36°	32°	12.5	N/A	0.6872	No collision
(b)	56°	15°	14.5	N/A	0.6279	No collision
(c)	36°	32°	12.2	0.6878	0.6857	Collision
(d)	56°	15°	14.2	0.6375	0.6370	Collision

CRediT authorship contribution statement

Hee-Jin Lee: Conceptualization, Methodology, Validation, Visualization, Investigation, Software, Writing – original draft, Investigation, Formal analysis. **Yoshitaka Furukawa:** Conceptualization, Resources, Validation, Writing – review & editing, Supervision, Formal analysis, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

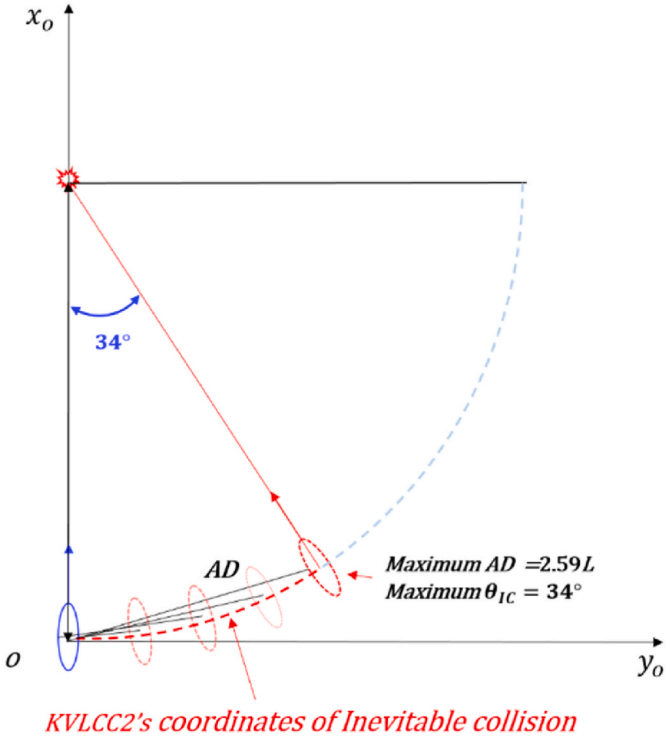


Fig. 18. Coordinates defining the inevitable collision area between two KVLCC2 ship models sailing at the same speed.

References

Ahn, J.-H., Rhee, K.-P., You, Y.-J.A., 2012. Study on the collision avoidance of a ship using neural networks and fuzzy logic. Appl. Ocean Res. 37, 62–173.

Chai, T., Weng, J., De-qi, X., 2017. Development of a quantitative risk assessment model for ship collisions in fairways. Saf. Sci. 91, 71–83.

Cockcroft, A.N., Lameijer, J.N.F., 2003. Guide to the Collision Avoidance Rules. Elsevier.

Coldwell, T.G., 1983. Marine traffic behaviors in restricted waters. J. Navig. 36 (3), 430–444.

COLREGS, 1972. Convention on the International Regulations for Preventing Collisions at Sea 1972 (With Amendments to 2007 in Force from 1 December 2009). International Maritime Organization.

Davis, P.V., Dove, M.J., Stockel, C.T., 1980. A computer simulation of marine traffic using domains and arenas. J. Navig. 33 (2), 215–222.

Du, L., Goerlandt, F., Banda, O.A.V., Huang, Y., Wen, Y., Kujala, P., 2020a. Improving stand-on ship's situational awareness by estimating the intention of the give-way ship. Ocean. Eng. 201, 107110.

Du, L., Banda, O.A.V., Goerlandt, F., Huang, Y., Kujala, P., 2020b. A COLREG-compliant ship collision alert system for stand-on vessels. Ocean. Eng. 218, 107866.

Fujii, Y., Tanaka, K., 1971. Traffic capacity. J. Navig. 24 (4), 543–552.

Gao, M., Shi, G.Y., 2020. Ship collision avoidance anthropomorphic decision-making for structured learning based on AIS with Seq-CGAN. Ocean. Eng. 217, 107922.

Gil, M., Montewka, J., Krata, P., Hinz, T., Hirdaris, S., 2020. Determination of the dynamic critical maneuvering area in an encounter between two vessels: operation with negligible environmental disruption. Ocean. Eng. 213, 107709.

Goerlandt, F., Kujala, P., 2011. Traffic simulation based ship collision probability modeling. Reliab. Eng. Syst. Saf. 96 (1), 91–107.

Goodwin, E.M., 1975. A statistical study of ship domains. J. Navig. 28 (3), 328–344.

Hasegawa, K., 1989. Fuzzy modelling of the behaviors and decision-making of ship navigators. In: Proc. Of 3rd International Fuzzy Systems Association (IFSA) Congress, pp. 663–666.

Hasegawa, K., Kouzuki, A., 1987. Automation collision avoidance system for ships using fuzzy control. J. Kanasai Soc. Naval Architect. 1–10.

He, Y., Jin, Y., Huang, L., Xiong, Y., Chen, P., Mou, J., 2017. Quantitative analysis of COLREG rules and seamanship for autonomous collision avoidance at open sea. Ocean. Eng. 140, 281–291.

Heij, C., Bijwaard, G.E., Knapp, S., 2011. Ship inspection strategies: effects on maritime safety and environmental protection. Transport. Res. Transport Environ. 16 (1), 42–48.

Hilgert, H., Baldauf, M., 1997. A common risk model for the assessment of encounter situations on board ships. Deutsche Hydrografische Zeitschrift 49 (4), 531–542.

Hörteborn, A., Ringsberg, J.W., 2021. A method for risk analysis of ship collisions with stationary infrastructure using AIS data and a ship manoeuvring simulator. Ocean. Eng. 235, 109396.

Hörteborn, A., Ringsberg, J.W., Svanberg, M., Holm, H., 2019. A revisit of the definition of the ship domain based on AIS analysis. J. Navig. 72 (3), 777–794.

Im, N., Luong, T.N., 2019. Potential risk ship domain as a danger criterion for real-time ship collision risk evaluation. Ocean. Eng. 194, 106610.

Kijima, K., Furukawa, Y., 2001. Design of automatic collision avoidance system using fuzzy inference. IFAC Proceedings 34 (7), 65–70.

Kijima, K., Furukawa, Y., 2002. Development of collision avoidance algorithm using fuzzy inference. In: Proceedings of the Fifth ISOPE Pacific/Asia Offshore Mechanics Symposium, pp. X123–X130.

Kijima, K., Katsuno, T., Nakiri, Y., Furukawa, Y., 1990. On the manoeuvring performance of a ship with the parameter of loading condition. J. Soc. Nav. Archit. Jpn. 168, 141–148.

Lenart, A.S., 2015. Analysis of collision threat parameters and criteria. J. Navig. 68 (5), 887–896.

Li, B., Pang, F.W., 2013. An approach of vessel collision risk assessment based on the D–S evidence theory. Ocean. Eng. 74, 16–21.

Li, S., Liu, J., Negenborn, R.R., 2019. Distributed coordination for collision avoidance of multiple ships considering ship maneuverability. Ocean. Eng. 181, 212–226.

Liu, J., Zhou, F., Li, Z., Wang, M., Liu, R.W., 2016. Dynamic ship domain models for capacity analysis of restricted water channels. J. Navig. 69 (3), 481–503.

Mou, J.M., Van Der Tak, C., Ligteringen, H., 2010. Study on collision avoidance in busy waterways by using AIS data. Ocean. Eng. 37 (5–6), 483–490.

Park, J., Kim, J., 2016. Predictive evaluation of ship collision risk using the concept of probability flow. IEEE J. Ocean. Eng. 42 (4), 836–845.

Park, S., Kim, B., Choi, B., Kim, E., Lee, H., Kang, H.J., 2011. A soft computing approach for collision risk assessments. In: 2011 11th International Conference on Control. Automation and Systems, pp. 1908–1910.

Perera, L.P., Carvalho, J.P., Soares, C.G., 2012. Intelligent ocean navigation and fuzzy-Bayesian decision/action formulation. IEEE J. Ocean. Eng. 37 (2), 204–219.

Pietrzykowski, Z., 2008. Ship's fuzzy domain–A criterion for navigational safety in narrow fairways. J. Navig. 61 (3), 499–514.

SOLAS, 1974. International Convention for the Safety of Life at Sea. International Maritime Organization.

Szlapczynski, R., Szlapczynska, J., 2017. Review of ship safety domains: models and applications. Ocean. Eng. 145, 277–289.

Szlapczynski, R., Krata, P., Szlapczynska, J., 2018. A ship domain-based method of determining action distances for evasive manoeuvres in stand-on situations. J. Adv. Transport. 2018, 1–19, 2018.

Wu, B., Yip, T.L., Yan, X., Soares, C.G., 2019. Fuzzy logic based approach for ship-bridge collision alert system. Ocean. Eng. 187, 106152.

Yasukawa, H., Yoshimura, Y., 2015. Introduction of MMG standard method for ship maneuvering predictions. J. Mar. Sci. Technol. 20 (1), 37–52.

Yoshimura, Y., 2005. Mathematical model for manoeuvring ship motion (MMG Model). In: Workshop on Mathematical Models for Operations Involving Ship-Ship Interaction, pp. 1–6.

Zhang, J., Yan, X., Chen, X., Sang, L., Zhang, D., 2012. A novel approach for assistance with anti-collision decision making based on the International Regulations for Preventing Collisions at Sea. Proc. IME M J. Eng. Marit. Environ. 226 (3), 250–259.

Zhang, W., Goerlandt, F., Montewka, J., Kujala, P., 2015. A method for detecting possible near miss ship collisions from AIS data. Ocean. Eng. 107, 60–69.

Zhang, W., Goerlandt, F., Kujala, P., Wang, Y., 2016. An advanced method for detecting possible near miss ship collisions from AIS data. Ocean. Eng. 124, 141–156.

Zhang, W., Kopca, C., Tang, J., Ma, D., Wang, Y., 2017. A systematic approach for collision risk analysis based on AIS data. J. Navig. 70 (5), 1117–1132.

Zheng, Z., Wu, Z., 2000. Research on Automatic Decision-Making System of Vessel Collision Avoidance. Dalian Maritime University, pp. 13–25.

Zhao, J., Wu, Z., Wang, F., 1993. Comments on ship domains. J. Navig. 46 (3), 422–436.

Zheng, K., Chen, Y., Jiang, Y., Qiao, S., 2020. A SVM based ship collision risk assessment algorithm. Ocean. Eng. 202, 107062.